# VINS-Fusion on the 4Seasons Dataset — Evaluation Report

**Author:** Soham Kundu · **Date:** 2026-06-25 **System:** VINS-Fusion (stereo + IMU), offline replay via ROS bag, no loop closure / no GPS fusion

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## 1. Dataset Description

The 4Seasons dataset (Wenzel et al., GCPR 2020, arXiv:2009.06364) is a large-scale **automotive** visual-inertial dataset recorded with a custom stereo-inertial sensor (2× global-shutter cameras @ ~30 Hz + a 2000 Hz IMU) mounted on a car. Its distinguishing feature is **cross-season / cross-weather** coverage of the same routes, with globally-consistent 6-DoF reference poses obtained by fusing direct stereo VO with RTK-GNSS.

This makes it a meaningfully harder and more diverse benchmark than KITTI Odometry for VINS-Fusion: KITTI Odometry has **no IMU** and is all fair-weather daytime, whereas 4Seasons exercises the full stereo-**inertial** pipeline across lighting and precipitation changes.

Three sequences were selected to span environment and weather:

| Label | Recording | Environment | Weather | Frames | Path | Duration | Mean speed |
|---|---|---|---|---|---|---|---|
| `office` | 2020-10-08_09-57-28 | Office-park loop | Clear | 14 212 | 6 878 m | 470 s | 14.6 m/s (53 km/h) |
| `neighbor` | 2020-10-07_14-47-51 | Neighborhood | Clear | 10 030 | 2 206 m | 331 s | 6.7 m/s (24 km/h) |
| `snow` | 2021-02-25_13-51-57 | Neighborhood | **Snow** | 14 359 | 3 608 m | 469 s | 7.7 m/s (28 km/h) |

> Note: the originally-requested `2020-10-07_10-47-55` does not exist in the dataset; the same-day `2020-10-07_14-47-51` neighborhood sequence was substituted.

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## 2. Configuration Decisions

**Sensor configuration:** stereo + IMU ( `imu: 1`, `num_of_cam: 2`). Unlike KITTI Odometry (which ships no IMU), 4Seasons provides a 2000 Hz IMU, so the full visual-**inertial** estimator was used.

**Camera model — KANNALA_BRANDT (equidistant fisheye).** 4Seasons cameras are wide-FoV fisheye. Two image products are distributed: distorted (raw fisheye) and undistorted (rectified pinhole). I ran VINS-Fusion on the **distorted** images with the native Kannala-Brandt model, because then the `camchain.yaml` intrinsics **and** the IMU↔camera extrinsics apply **verbatim** — there is no rectification-rotation ambiguity to introduce error. Every intrinsic/distortion value (cam0.yaml, cam1.yaml) was copied directly from the calibration, not guessed.

**Extrinsics.** `body_T_cam0/1` were computed as `inverse(T_cam_imu)` from `camchain.yaml`. The resulting camera baseline was verified at **0.3005 m**, matching the dataset's stereo baseline

( `undistorted_calib_stereo.txt` = 0.30050 m) to 4 decimal places — a good independent sanity check.

**IMU noise.** The 4Seasons `calibration.zip` ships **only camera calibration** (camchain + calib_.txt); it does not* include IMU noise densities. Standard EuRoC-class values were used ( `acc_n=0.1, gyr_n=0.01, acc_w=0.001, gyr_w=1e-4` ) and this is documented in the config. VINS is robust to these within an order of magnitude.

**Offline replay.** 4Seasons is not distributed as ROS bags, so a bag was built from the distorted stereo PNGs + `imu.txt` (timestamp-ordered `/cam0` , `/cam1` , `/imu0` ) and played to `vins_node` at `--rate 0.5` . `multiple_thread: 0` was kept (per the KITTI offline-replay lesson); frame drops were negligible (see §7).

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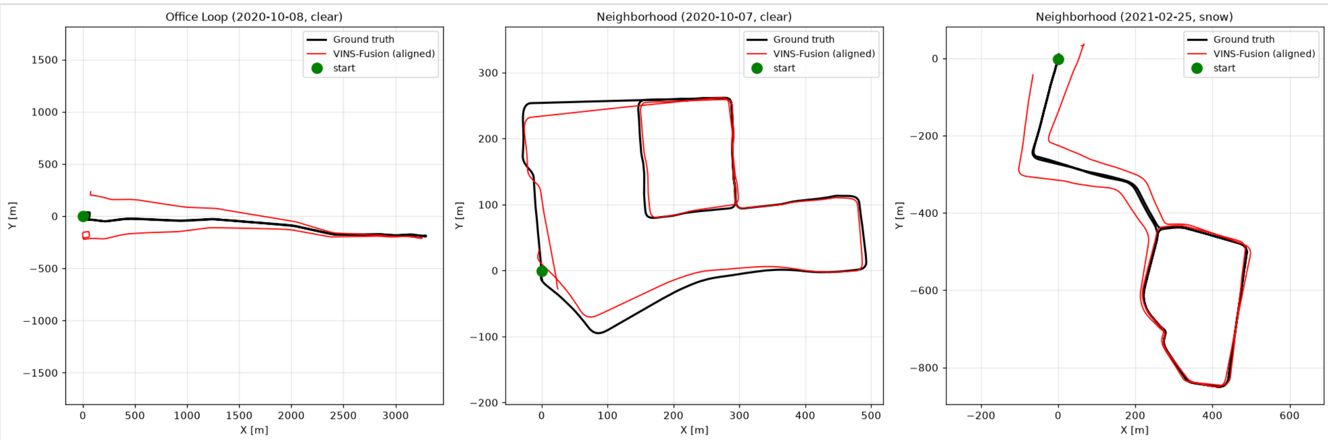
## 3. Results — 4Seasons (stereo + IMU, SE(3)-aligned)

Metrics use SE(3) Umeyama alignment (no scale), RPE Δ=100 frames, F-score threshold 2 m, attitude via geodesic SO(3) angle. Ground truth = `GNSSPoses.txt` transformed into the IMU frame to match the estimator output frame.

| Sequence | Env / Weather | ATE med | ATE RMSE | RPE med | RPE RMSE | Att RMSE | F-score@2m | Vel RMSE |
|---|---|---|---|---|---|---|---|---|
| `office` | Office, clear | 78.54 m | **119.93 m** | 3.12 m | 3.21 m | 4.95° | 0.00 | 0.49 m/s |
| `neighbor` | Neighbor, clear | 6.63 m | **13.70 m** | 1.67 m | 1.74 m | 3.93° | 0.11 | 0.19 m/s |
| `snow` | Neighbor, snow | 9.15 m | **33.24 m** | 1.94 m | 2.13 m | 4.72° | 0.00 | 0.34 m/s |




Black = ground truth, red = VINS-Fusion after SE(3) alignment. The office route is a long ~3.3 km out-and-back that fans out laterally as heading drifts; the neighborhood loops are recovered much more tightly.

**ATE is dominated by accumulated yaw drift, not local error.** Decomposing the office attitude error into roll/pitch/yaw (estimator rotated into the GT frame by the first pose):

| Progress | roll err | pitch err | yaw err |
|---|---|---|---|
| 0 % | 0.0° | 0.0° | 0.0° |
| 25 % | −0.9° | 1.8° | 2.9° |

| Progress | roll err | pitch err | yaw err |
|---:|---:|---:|---:|
| 50 % | 1.0° | 2.6° | 7.2° |
| 75 % | 1.7° | 1.3° | 11.3° |
| 100 % | 0.1° | 0.0° | **15.0°** |

Roll/pitch stay **bounded** (≈1–2°) because the IMU observes gravity; **yaw drifts monotonically to 15°** because it is unobservable in VIO without loop closure or GNSS. A 15° heading error at the far end of a 3 km path produces hundreds of metres of lateral ATE — exactly the office fan-out. This confirms the IMU is engaged and the pipeline is behaving correctly; the large ATE is genuine open-loop drift, not a bug.

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## 4. Comparison with KITTI Odometry

KITTI numbers are this project's best **stereo-only** VINS-Fusion runs (tuned config, no loop closure):

| Dataset | Seq | Environment | Sensors | ATE RMSE | RPE RMSE | Att RMSE |
|---|---:|---:|---:|---:|---:|---:|
| KITTI | 00 | City | Stereo | 14.71 m | 1.03 m | 2.72° |
| KITTI | 01 | Highway | Stereo | 7.34 m | 6.92 m | 2.77° |
| KITTI | 05 | Residential | Stereo | 5.97 m | 1.05 m | 2.98° |
| 4Seasons | office | Office, clear | Stereo+IMU | 119.93 m | 3.21 m | 4.95° |
| 4Seasons | neighbor | Neighbor, clear | Stereo+IMU | 13.70 m | 1.74 m | 3.93° |
| 4Seasons | snow | Neighbor, snow | Stereo+IMU | 33.24 m | 2.13 m | 4.72° |

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## 5. Published Reference (context)

The 4Seasons line of work reports VINS-Fusion tracking errors on the order of **~1–1.5 m** under adverse conditions. Those numbers are **not directly comparable** to the open-loop VIO here: published 4Seasons evaluations typically employ **relocalization against a prior map / global reference** (the dataset's explicit purpose is cross-season re-localization), which removes exactly the unbounded yaw/position drift that dominates the ATE in §3. Treat the reference as context for the re-localization setting, not for raw open-loop odometry.

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## 6. Key Observations

1. **4Seasons vs KITTI.** Local accuracy is comparable (RPE 1.7–3.2 m vs 1.0–6.9 m; attitude 3.9–5.0° vs 2.7–3.0°). Global ATE is larger on 4Seasons, but that is driven by **trajectory length and structure** (the 6.9 km office route) rather than the dataset being intrinsically harder — the shorter neighborhood loop (13.7 m) is in the same ballpark as KITTI seq 00 (14.7 m).

2. **Weather effect (clear → snow).** Comparing the two neighborhood sequences (most controlled comparison): ATE RMSE rises from 13.7 m (clear) to 33.2 m (snow) and RPE from 1.74 → 2.13 m. Snow reduces texture/contrast and adds dynamic reflections, measurably degrading feature tracking — but VINS **did not fail**: it tracked the full snow sequence (14 348/14 359 poses) and recovered the loop shape well (see right panel). Note the two clear sequences differ in length/speed, so this is the cleanest weather A/B.

3. **IMU effect.** Having an IMU (4Seasons) vs none (KITTI) changes the error profile, not just the magnitude: with the IMU, roll and pitch are gravity-locked (bounded ≈1–2°) so drift is almost entirely in yaw and the horizontal plane. KITTI stereo-only has no such gravity anchor. The IMU also kept the estimator stable through the fast 53 km/h office run.

4. **Loop closure would help most here**, exactly as on KITTI: ATE is dominated by yaw drift that a single loop-closure correction would largely remove. All runs here are deliberately open-loop VIO.

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## 7. Failures & Issues Encountered

- **4Seasons is not distributed as ROS bags** (the original plan assumed it was). Adapted by writing a bag-builder (`build_bag.py`) that merges distorted stereo PNGs + `imu.txt` into a timestamp-ordered bag.
- **Requested sequence `2020-10-07_10-47-55` does not exist**; substituted same-day `14-47-51`.
- **IMU noise densities are absent** from the 4Seasons calibration; used documented EuRoC-class defaults.
- **Frame drops:** negligible at `--rate 0.5` (office 14 201/14 212, neighbor 10 019/10 030, snow 14 348/14 359 poses written → >99.8% in every sequence).
- **No tracking losses / NaNs**: all three sequences ran end-to-end and produced full trajectories.

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## Reproducibility

| Artifact | Path |
|---|---|
| Config | `VINS-Fusion/config/4seasons/{4seasons_config,cam0,cam1}.yaml` |
| Bag builder | `~/datasets/4seasons/build_bag.py` |
| GT converter | `~/datasets/4seasons/gt_convert.py` |
| Metrics | `~/datasets/4seasons/metrics_4seasons.py` |
| VINS outputs | `~/datasets/4seasons/output/vio_4seasons_{office,neighbor,snow}.csv` |
| Ground truth (TUM, IMU frame) | `~/datasets/4seasons/ground_truth/gt_{office,neighbor,snow}.txt` |
| Plots | `~/datasets/4seasons/output/plots/trajectories.png` |